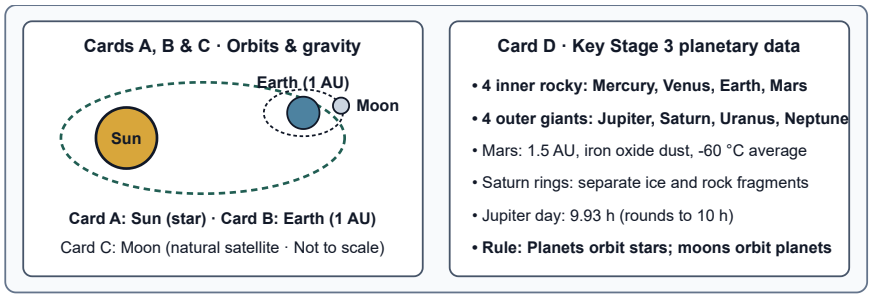
Reference sheets

Reference A · Class Solar System Source Cards (Cards A, B, C, and D)

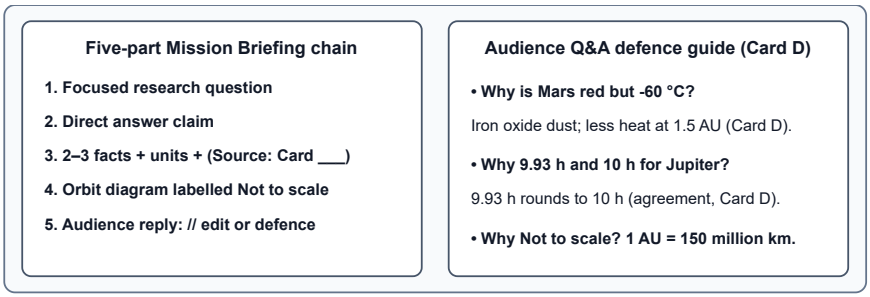


Class Source Cards A, B, C, and D showing the Sun's gravity, Earth's 1 AU orbit, the Moon's satellite orbit, and Key Stage 3 planetary data.

Reference only. Do not write on this sheet. Use these verified cards to build your Mission Briefing and answer audience questions.

Source card code	Body and classification	Verified science facts on this card	Key Stage 3 mechanism or measurement
Card A · Star card	The Sun (central star)	The Sun is a hot star at the centre of our Solar System that emits its own light	The Sun's non-contact gravitational pull holds planets in curved orbits around it
Card B · Planet card	Earth (inner rocky planet)	Earth is a nearly spherical rocky planet that orbits the Sun once a year	Earth orbits at an average distance of 1 astronomical unit (1 AU, about 150 million km)
Card C · Moon card	The Moon (natural satellite)	The Moon is a rocky natural satellite that orbits Earth in about a month and reflects sunlight	Earth's gravity holds the Moon in orbit around Earth rather than directly around the Sun
Card D · Solar System data	4 inner rocky planets & 4 outer giants	Inner rocky: Mercury, Venus, Earth, Mars. Outer giants: Jupiter, Saturn, Uranus, Neptune	Mars (1.5 AU, -60 °C, iron oxide dust); Saturn rings (ice and rock fragments); Jupiter day (9.93 h, rounds to 10 h)

Reference B · Five-part Mission Briefing and audience Q&A guide



Five-part Mission Briefing chain (Question, Claim, Facts + Card codes, Not to scale diagram, Audience reply) and Key Stage 3 defence guide.

Briefing part	What to include in your presentation	Example from Cards A, B, C, and D
1. Research question	One focused question about a Solar System body or pattern	'Why is Mars colder than Earth even though Mars looks red?'
2. Answer claim	One direct scientific sentence that answers your question	'Mars is colder because it orbits further from the Sun, and its red colour comes from rusty surface dust.'
3. Supporting facts & codes	Two or three verified facts with units and (Source: Card ___)	'Earth orbits at 1 AU (Source: Card B); Mars orbits at 1.5 AU, averages -60 °C, and has iron oxide dust (Source: Card D).'
4. Diagram & scale note	A clear diagram of the bodies/orbits plus the words 'Not to scale'	'Sun, Earth (1 AU), and Mars (1.5 AU) orbit diagram — Not to scale (1 AU = 150 million km).'
5. Audience Q&A response	Answer a listener's question using a card code, then mark // or defend	'9.93 h and 10 h for Jupiter's day agree because 9.93 h rounds to 10 h (Source: Card D).'

Made by Matt

Tier pack · Independent

Use Reference A and Reference B. Work in your book or on this sheet. Point, speak, sign, draw, type, write, or tell an adult. You may pass.

Arrival · 4 questions

1. Use the arrival picture cue. What three steps turn a research note into a complete mission briefing?

Help: Read the headings across the left, middle, and right panels of the drawing.

2. Why does Step 2 in the arrival picture cue include the label 'Not to scale'?

Help: Compare the real distance between the Sun and Earth with the size of the box.

3. In Step 3 of the arrival picture cue, how does the presenter respond to the audience question 'Which card backs the Moon fact?'

Help: Check the response box in the right panel.

4. Why is a focused research question (such as 'Why is Mars cold and red?') better for a short briefing than 'Tell me everything about space'?

Help: Think about whether a one-minute briefing can cover every object in the universe.

You do tasks

1. Use Reference A and Reference B to write a structured Solar System Mission Briefing with: (1) one focused research question, (2) one direct answer claim, and (3) two or three supporting facts with units and exact source codes (Card A, Card B, Card C, or Card D).

Staff I ☐ S ☐ C ☐

Help: Keep every fact linked to your question and include units (such as 1 AU, 1.5 AU, -60 °C, or 9.93 hours) plus (Source: Card ___).

2. Explain what your Mission Briefing diagram helps the audience see, and state why you must include the label 'Not to scale'. **Staff I ☐ S ☐ C ☐**

Help: Compare the orbit order shown in your drawing with the real 150 million km (1 AU) distance on Reference A.

3. Respond to this audience question using Reference A and Reference B: 'Your briefing says Saturn has rings — could a spacecraft land on Saturn's rings as if they were a solid bridge?' Cite the card and show your // addition or defence. **Staff I ☐ S ☐ C ☐**

Help: Check the Saturn row on Card D of Reference A.

Optional participation-loop prompt: Present one part of your briefing to an adult, hear one audience question or feedback step, and respond below or with // in your book. (Staff: the pupil responds to feedback with an adult, and R is marked only after the pupil has responded.)

Exit ticket

1. State your research question, your answer claim, and one verified fact with its unit and source card code.

2. Why must a classroom diagram of the Sun, Earth (1 AU), and Moon include the label 'Not to scale'?

3. Explain how you answered the audience question and what you changed with // or defended using Cards A to D.

Pupil voice

What helped you today, and what would you change?

Point, speak, sign, draw, write or choose. You may pass.

Feedback and response · Independent

Feedback codes

Staff I ☐ S ☐ C ☐ · Route card feedback boxes: VF ☐ · WS ☐ · I ☐ · E ☐ · NS + … ☐ · R ☐ · // ☐ · ? ☐

Code	Meaning
VF	Verbal feedback
WS	Worked with support
I	Independent
NS + …	Next step
E	Evidence captured
R	Pupil responded to feedback
//	Self-edit point
?	Read this back to me

My response

My response: Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt